Md. A. B. M. Saif Ahmed Sany

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Professional Summary

M.Sc. Materials Engineering (Werkstofftechnik) student at RWTH Aachen with 2+ years of experience in product development, manufacturing operations, and industrial projects, including taking consumer electronics from market research and concept through prototype validation. Skilled in SolidWorks (CSWA-certified), AutoCAD, Oracle EBS, Excel, Python, and technical documentation, with presentation experience across industries and conferences. Seeking a Werkstudent position in manufacturing or product development.

Education

RWTH Aachen University - Aachen, Germany Oct 2026 - Present
Master of Science in Materials Engineering (Werkstofftechnik)

Khulna University of Engineering and Technology - Khulna, Bangladesh Feb 2017 - May 2024
Bachelor of Science in Mechanical Engineering

Professional Experience

Samuda Chemical Complex Limited – Dhaka, Bangladesh Feb 2026 – Sep 2026
Mechanical Engineer | Project Management Division

- Integrated **vendor-supplied 2D CAD drawings** into the overall plant layout for an FRP rebar and brick factory; **prepared civil and mechanical BOQs** for construction and machinery installation and supported contractor coordination and material-requisition reviews.
- Evaluated **vendor quotations and technical specifications** for industrial machinery and heavy equipment, including excavators, cranes, vibratory hammers, and forklifts, to assess suitability and support purchasing decisions.
- Contributed to preliminary **feasibility and market studies** for a proposed **steel pipe pile manufacturing** joint venture with **Mitsui & Co. Asia Pacific Pte. Ltd. and JFE Steel**, preparing technical presentations on market demand, applications, and product portfolios.
- Developed **technical proposals and capacity calculations** for marine infrastructure projects, including a grain terminal with jetty and bulk storage and a multi-product oil jetty; coordinated grain-terminal feasibility/design information with **Nippon Koei** and issued civil-construction RFQs to **CRCC and CHEC** for the oil jetty.

PRAN - RFL Group – Dhaka, Bangladesh Oct 2024 – Feb 2026
Assistant Manager | Production & Operation (*Domestic Water Heater*)

- Led **product development** from concept through physical testing for **5+ injection-molded, thermoformed, sheet-metal, and die-cast parts and assemblies**, benchmarked against Bosch, Stiebel Eltron, and A. O. Smith; used **Gantt charts** to coordinate development, design, and sourcing timelines.
- Managed **demand-based production planning** and daily performance tracking, achieving **100% of KPIs** for production, inventory utilization, rejection rate, and HR cost.
- Improved **manufacturing efficiency by 12%** through production-process and machinery optimization; reduced **material waste by up to 15%** through consumption and variance analysis.
- **Automated production reporting, OEE data entry, and raw-material stock monitoring** using Python, Google Apps Script, Google Sheets, and WhatsApp-based reporting workflows, supporting timely procurement and performance tracking.
- Analyzed **inventory movements** to identify slow-moving stock and implement corrective measures, **reducing carrying costs by 15%** and improving inventory turnover.
- Coordinated **procurement of spare parts, CKD items, and imported products** by analyzing supplier quotations, evaluating quality and delivery requirements, and supporting cost-saving

opportunities; **optimized ERP workflows** to improve order-fulfilment accuracy and reduce manual errors.

- Led **Six Sigma and Kaizen** initiatives that increased production throughput by **15% and reduced production defects by 10%**, supporting continuous improvement in manufacturing operations.

Technical Skills

Microsoft Office	:	Word | Power Point | Excel
ERP & Automation	:	Oracle EBS | Google Apps Script | VBA Excel
Data Analysis	:	Power BI | Python | Ansys
CAD	:	SOLIDWORKS | AutoCAD
Language	:	English (IELTS -7.0 C1), German (Learning), Bangla (Native)

Projects

Product Development Project for Instant Water Heater Dec 2024 – June 2025

- Led development of new SKU by defining scope, creating WBS, and managing budgets to deliver new products on schedule.
- Managed design and manufacturing, overseeing material selection, supplier RFQs, BOM generation, and SOP development, ensuring quality and scalability.
- Led the market launch by establishing distribution channels and optimizing supply chains, achieving timely launches and continuous post-launch improvements.

Automated Reporting for better Inventory Management Jan 2025 – Feb 2025

- Developed an automated reporting system to streamline inventory analysis and optimize stock levels to support data-driven procurement decisions.

Development of a Wall Climbing Bot Sep 2019 – Dec 2019

- Constructed a robot capable of climbing vertical surfaces by Bernoulli's principle through use of negative pressure exerted on wall surface.
- Designed a CAD model portraying its visualization and performance analysis.

Publications

- *The Impact of ZnO Nanofiller on the Mechanical Characteristics of Epoxy Composites Reinforced with Nylon-Banana Fibers*, Journal of Engineering and Technology (JET).

Certification

- Certified SolidWorks Associate (CSWA)
- ISO 9001:2015 – Internal Auditor Training
- Six-Sigma Green Belt

Hobbies

Hiking, Watching Movies, Literature, Playing Guitar, Exploring new places.